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R180T variant of δ-ornithine aminotransferase associated with gyrate atrophy: biochemical, computational, X-ray and NMR studies provide insight into its catalytic features

Riccardo Montioli[1], Alessandro Paiardini[2], Giorgio Giardina[2], Serena Zanzoni[3], Francesca Cutruzzola[2], Barbara Cellini[4] and Carla Borri Voltattorni[1]

1 Department of Neurosciences, Biomedicine and Movement Sciences, University of Verona, Italy
2 Department of Biochemical Sciences "A. Rossi Fanelli", Sapienza University of Rome, Italy
3 Centro Piattaforme Tecnologiche, University of Verona, Italy
4 Department of Experimental Medicine, University of Perugia, Italy

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Correspondence
C. B. Voltattorni, Department of Neurosciences, Biomedicine and Movement Sciences, Section of Biological Chemistry, University of Verona, Strada Le Grazie, 8, 37134 Verona, Italy
Tel: +39 0458027175
E-mail: carla.borrivoltattorni@univr.it

Riccardo Montioli and Alessandro Paiardini contributed equally to this work.

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Among the over 50 gyrate atrophy-causing mutations of ornithine δ-aminotransferase (OAT), the R180T involves an active site residue located at the dimer interface, which in the crystal structure of OAT complexed with 5-fluoromethylornithine engages a salt bridge with the α-carboxylate of the substrate analogue. Starting from the previous finding that no transaminase activity was detected in CHO-K_1 cells expressing the R180T variant, here we try to shed light at the protein level on the structural and/or functional defects of the R180T variant. To this aim, the variant has been cloned, expressed, purified and characterized by a combination of biochemical and structural studies. Although the R180T variant shares a similar overall conformation with the wild-type, its crystal structure solved at 1.8 Å reveals slight structural alterations at the active site and at the dimeric interface. These changes are consistent with the spectroscopic and kinetic results, indicating that the variant, as compared with the wild-type OAT, shows (a) an increased K_m value for L-ornithine (L-Orn), (b) an altered pyridoxal 5′-phosphate binding mode and affinity and (c) an increased thermostability. In addition, the R180T mutant exhibits a remarkable loss of catalytic activity and is endowed with the ability to catalyse not only the δ-transamination but also, albeit to a lesser extent, the α-transamination of L-Orn. Overall, these data indicate that the slight structural changes caused by the R180T mutation, preventing a proper collocation of L-Orn at the active site of OAT, are responsible for the notable reduction of the catalytic efficiency.

Enzymes
Ornithine aminotransferase EC 2.6.1.13.

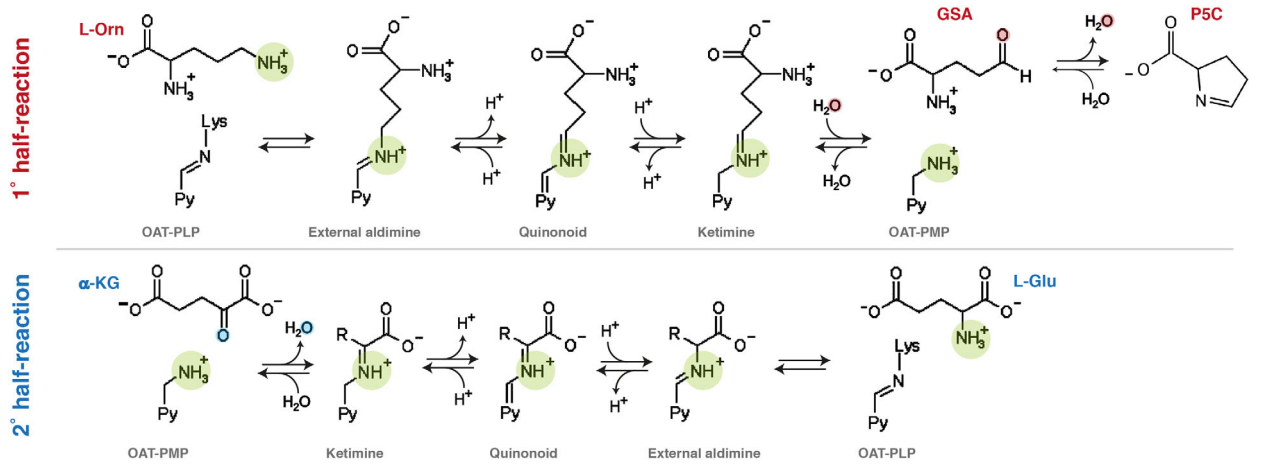
Databases
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Abbreviations
α-KG, α-ketoglutarate; 5FMO, 5-fluoromethylornithine; GA, gyrate atrophy; L-Orn, L-ornithine; P2C, 1-pyrroline-2-carboxylate; P5C, 1-pyrroline-5-carboxylate; PLP, pyridoxal 5′-phosphate; PMP, pyridoxamine 5-phosphate.

Introduction

Ornithine δ-aminotransferase (OAT; EC 2.6.1.13) is a mitochondrial enzyme requiring pyridoxal 5′-phosphate (PLP) as coenzyme. The enzyme converts L-ornithine (L-Orn) and α-ketoglutarate (α-KG) to glutamic-γ-semialdehyde (GSA) and L-glutamate respectively. The first product of this reaction undergoes spontaneous cyclization forming Δ^1-pyrroline-5-carboxylate (P5C). Like every PLP-dependent transaminases, OAT operates via a 'ping-pong' mechanism, in which two half-reactions complete a full transamination cycle. As shown in Scheme 1, in the first half-transamination reaction, L-Orn reacts with the PLP form of the enzyme (OAT-PLP) to yield GSA and the pyridoxamine 5′-phosphate form of the enzyme (OAT-PMP). In the second half-transamination, α-KG reacts with OAT-PMP to reform OAT-PLP and glutamate. A recent review summarizes current knowledge on OAT, in particular about its physical and chemical aspects, its synthesis and fate, the localization, the physiological role, the involvement of the enzyme in health and diseases as well as the structural features [1]. The human enzyme is synthesized on cytoplasmic free ribosomes as a 45 kDa precursor and is processed into a mature protein after mitochondrial entry [2]. OAT is expressed in many tissues, the highest level is found in liver, kidney, intestine, lung, endocrine tissues, brain and eyes [3]. Structure analyses revealed that OAT has a tetrameric assembly composed by two identical homodimers [4]. The crystal structure of ligand-free human recombinant OAT and its complex with the highly and potent inhibitor 5-fluoromethylornithine (PDB: 5FMO) was solved [5,6]. The overall structure shows a large similarity to other known PLP-dependent enzymes of subgroup II, i.e. glutamate-1-semialdehyde aminomutase (PDB: 2HOY), ω-amino: pyruvate aminotransferase (PDB: 5LH9) and dialkylglycine decarboxylase (PDB: 1D7S). Each OAT monomer is composed by a large domain of 249 residues that mostly contributes to the dimer interface, a small C-terminal domain of 95 residues, as well as an N-terminal arm of 42 residues. The cofactor PLP (a) binds the active site Lys292 through a Schiff base linkage (internal aldimine), (b) is located in the large domain at the subunit interface and (c) interacts with residues from both subunits of a dimer. The solved structures also reveal that the main residues responsible for the specific binding of the substrate analogue are Arg180 and Tyr55. While Arg180 contacts the α-carboxylate by a strong salt bridge, Tyr55 engages the α-amino group by a short hydrogen bond. It is likely that these anchoring points are the same involved in the L-Orn binding.

Ornithine δ-aminotransferase deficiency is responsible for gyrate atrophy (GA), an autosomal recessive disorder causing a progressive degeneration of the choroid and retina epithelium leading to blindness in young adults [7]. The biochemical hallmark of the disease is a marked elevation of plasma and urine levels of L-Orn [8]. Over 50 missense mutations associated with GA were identified so far. Among these, two involve residues located at the active site of the enzyme: Tyr55 and Arg180. The Y55H mutation was found in a hemizygous state [9], while the R180T mutation in a homozygous form [10]. The first mutation was shown to cause a low level of OAT mRNA and antigen in cultured fibroblasts [11]. On the other hand, although CHO-K_1 cells transfected with the wild-type or the R180T mutant enzymes revealed in both cases substantial amounts of OAT-hybridizing mRNA, those transfected with the R180T mutant had



Scheme 1. Mechanism of the L-Orn transamination catalysed by OAT.

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not detectable transaminase activity even at a high PLP concentration [12]. On these premises, we tried to define at protein level the structural and functional effects of the R180T mutation on human OAT.

A combination of biochemical, computational, NMR and X-ray studies was employed. Global analysis of the results reveals that the R → T replacement at position 180 of OAT, although not changing the overall conformation of the enzyme, causes (a) structural effects on L-Orn binding, leading to an increased value of the K_m for L-Orn, (b) structural effects at the dimeric interface, possibly responsible for the alteration of the state of the bound coenzyme and for the increased thermostability, (c) production, as predicted by molecular simulations and verified by NMR analysis, in addition to P5C, of 1-pyrroline-2-carboxylate (P2C), generated by α-transamination of L-Orn and (d) a 100-fold decrease in the k_{cat} value for L-Orn, possibly due to the inability of the substrate to be properly accommodated at the active site of the mutant.

Results and Discussion

Spectroscopic features of the R180T variant

Wild-type OAT and R180T variant had similar secondary structure profiles, as determined by far-UV CD experiments. This indicates that the mutation does not affect the secondary structure of the enzyme. They also had a comparable tertiary organization owing to their similar near-UV CD spectra and intrinsic fluorescence spectra (Fig. 1A,B). Furthermore, both wild-type and R180T mutant eluted at identical time-points in size exclusion chromatography, signifying the tetrameric organization of the proteins. Nevertheless, we noticed that the mutant shows a small decrease of the intensity of the absorbance and dichroic band at 420 nm with respect to the wild-type. Thus, in order to acquire information on the effect of the mutation on the coenzyme binding region, the fluorescence emission intensity of the $NaBH_4$-reduced R180T variant was monitored and compared with that of the wild-type enzyme. Upon excitation at 336 nm, the wild-type exhibits an emission fluorescence spectrum centred at 385 nm, while the reduced mutant displays an intensity of the emission fluorescence about 1.5-fold lower and an emission maximum ~ 5 nm blue shifted compared to that of the wild-type OAT (Fig 2). The altered state of the bound coenzyme and of its microenvironment is corroborated by the fact that we were not able to obtain the apo form of the mutant. In fact, upon reaction of the R180T mutant with phenylhydrazine under the same experimental conditions already used for the wild-type OAT

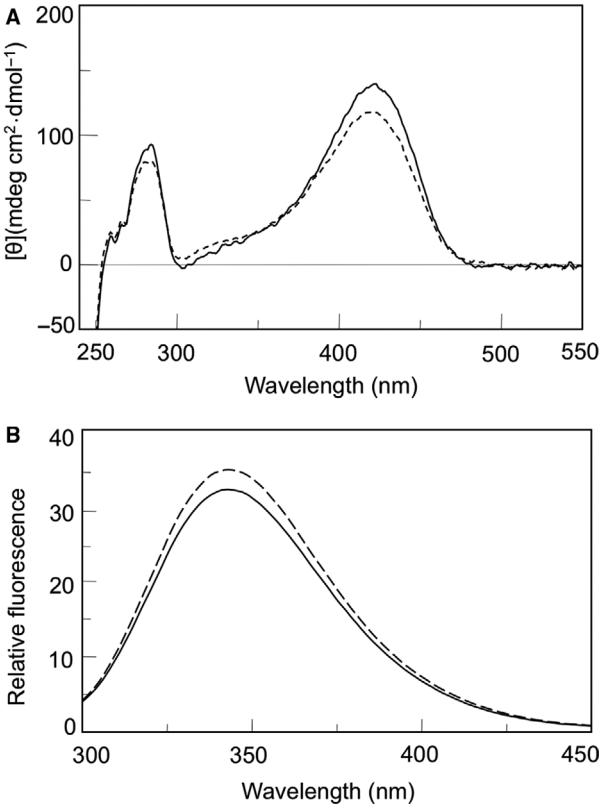


Fig. 1. CD and intrinsic fluorescence spectra of wild-type and R180T mutant OAT enzymes. (A) CD spectra of wild-type (–) and R180T variant (- - -) at an enzyme concentration of 6 μM. (B) Intrinsic fluorescence spectra (exc. at 280 nm) at enzyme concentration of 6 μM. Symbols as in (A). For all spectra, the buffer was 50 mM Hepes containing 150 mM NaCl, pH 8.

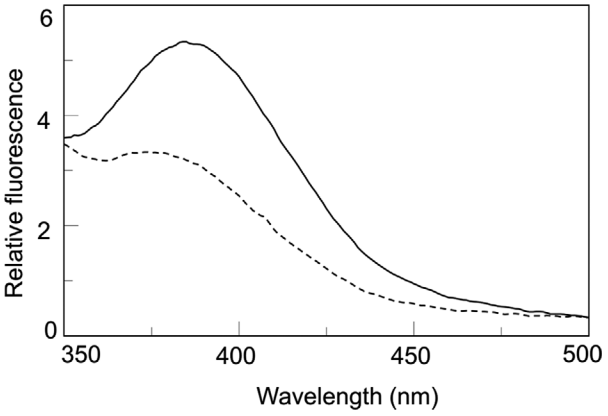


Fig. 2. Coenzyme fluorescence emission spectra of NaBH4-reduced wild-type and R180T OAT variant. The excitation wavelength was at 336 nm, the protein concentration was 10 μM. Symbols and buffer as in Fig. 1.

[4], no PLP release occurs. The same result was obtained when, instead of phenylhydrazine, hydroxylamine was used. An attempt to release PLP after 2-h incubation of

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R180T mutant with phenylhydrazine or hydroxylamine followed by dilution up to 100 nM enzyme concentration was also unsuccessful. Again, no apo form could be obtained after reaction of the R180T mutant with L-Orn followed by dialysis against 1 M potassium phosphate buffer, pH 6, in the presence of pyridoxamine. Finally, only 30% of the bound-PLP was released after 2 h incubation with phenylhydrazine followed by dilution up to 20 nM enzyme concentration. Altogether, these data strongly suggest that replacing Arg180 with Thr modifies the PLP binding mode and affinity. This is not surprising considering that Arg180 is located at the active site of the enzyme.

Thermal stability of the R180T variant

The thermostability of the R180T variant, either in the PLP or in the PMP form, was measured by CD-monitored thermal unfolding and compared with that of the wild-type enzyme. These studies were performed by monitoring the decrease of the dichroic signal at 220, 420 and 330 nm, which are indicative of the loss of the protein secondary structure, of the PLP-bound content and of the PMP-bound content respectively. All the CD-monitored heating scans reveal that, like the wild-type OAT, the mutant in the PLP or PMP form is denatured and releases PLP or PMP in a single-step concerted process. The data reported in Table 1 indicate that (a) the release of the PLP and the loss of the secondary structure of the wild-type and mutant OAT-PLP-bound enzymes occur with melting temperatures similar to each other (67 °C for the wild-type and 72 °C for the mutant), (b) the wild-type and mutant PMP-bound enzymes exhibit similar $T_{m(222\ nm)}$ and $T_{m(333\ nm)}$ values, even if lower than those of OAT enzymes in PLP form and (c) the R180T mutation appears to exert a stabilizing effect more pronounced for the PMP form (~ 12 °C) than for the PLP form (~ 5 °C). These results indicate that the PLP- and PMP-binding active sites as well as the secondary structures of the R180T mutant exhibit a higher thermal stability than the corresponding ones of the wild-type OAT.

Table 1. $T_{m(222\ nm)}$, $T_{mPMP(333\ nm)}$ and $T_{mPLP(420\ nm)}$ values of wild-type and R180T mutant OAT enzymes.

Enzyme	$T_{m(222\ nm)}$ (°C)	$T_{m(333\ nm)}$ (°C)	$T_{m(420\ nm)}$ (°C)
WT-PLP	67.0 ± 0.1		66.8 ± 0.1
R180T-PLP	72.6 ± 0.1		70.6 ± 0.3
WT-PMP	48.8 ± 0.1	50.1 ± 0.4	
R180T-PMP	60.1 ± 0.1	61.5 ± 0.5	

Steady-state kinetic parameters of the R180T variant

The steady-state kinetic parameters of the R180T variant for the L-Orn/αKG pair, together with those of the wild-type OAT, are reported in Table 2. The data indicate that the k_{cat} is reduced by about 100-fold, while the K_m values for L-Orn and α-KG increase by ~ 35- and 3-fold, respectively. The finding that the R180T mutation increases remarkably the K_m value for L-Orn is not surprising. In fact, Arg180 participates to substrate binding forming a strong salt bridge with the α-carboxylate of 5FMO, as observed from the structure of OAT formed upon reaction of the enzyme with the inhibitor 5FMO (PDB: 2OAT) [6] (Fig. 3A). Instead, it is not easy to understand the reason for the marked drop in the k_{cat} value. A possible explanation could come from an improper positioning of the side chain of L-Orn due to the structural changes of the PLP binding site. The latter, according to the Dunathan's hypothesis [13], could in turn determine a lower k_{cat} and/or the generation of byproducts of the reaction. To assess this hypothesis, X-ray, molecular docking and NMR studies were employed.

Crystal structure of the R180T variant

To shed light on the structural effects of the R180T mutation, we solved the OAT-R180T structure by X-ray crystallography at 1.8 Å resolution. The mutated side chain was clearly visible in the electron density maps (Fig. 3B) and model building was straightforward. The oligomeric state of the enzyme is unaffected by the mutation and the structure of OAT-R180T is virtually identical to the wild-type (PDB: 1OAT [3]). The backbone atoms superpose with a r.m.s.d of 0.25 Å (Fig. 3A), in line with CD spectroscopy data indicating the absence of gross structural alterations induced by the mutation. However, the replacement of Arg with Thr affects to a small extent the position of the main chain of the other monomer in the region His319′-Gly320′-Ser321′. In particular, the Cα of Gly320′ is shifted of 1.4 Å towards the active site (Fig. 3E). This is likely due to a relaxation of the backbone induced by the lack of the Arg bulky side chain. This shift renders the active site suboptimal for substrate binding, given that the main chain in the relaxed position would make a partial steric overlap with both oxygen atoms of the carboxyl group of Orn (Fig. 3F). It should be also noted that Arg180 is located at the dimer interface and forms an H-bond with the main-chain CO of Gly317′, belonging to the other subunit. At the same time the CO of Arg180

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Table 2. Steady-state kinetic parameters of wild-type and R180T mutant OAT enzymes.

Enzyme	Substrate	Co-substrate	k_{cat} (s^{-1})	K_m (mM)	k_{cat}/K_m ($mM^{-1}\cdot s^{-1}$)
Wild-type	L-Orn	α-KG	34.5 ± 0.6	6.5 ± 0.4	5.3 ± 0.3
	α-KG	L-Orn	35.7 ± 0.8	3.9 ± 0.5	9.1 ± 1.2
R180T	L-Orn	α-KG	0.37 ± 0.01	237 ± 10	0.0016 ± 0.0001
	α-KG	L-Orn	0.41 ± 0.01	11.3 ± 0.6	0.037 ± 0.002

main chain is involved in an H-bond with the amino group of Lys151′, also belonging to the other subunit. The introduced mutation, while abolishing the interaction with Gly317′, reinforces the one with Lys151′, which now makes a bi-dentate H-bond with the main-chain CO and with the OH of the mutated side chain (Fig. 3D). These new interactions, coupled with the entropically favoured substitution of a bulky and flexible side chain with a small one, may have a stabilizing effect on the dimeric interface, resulting in the observed higher thermal stability of the mutant. The higher affinity for PLP and PMP may also be interpreted as a consequence of the higher interface stability. This effect has been shown for other PLP-dependent transferases [13–16], for which higher stability of the oligomeric interface correlates with higher affinity for PLP and vice versa.

Docking results

In order to identify the reason underlying the remarkable catalytic consequence of the Arg180 amino acid substitution with Thr, a molecular docking mimicking the L-Orn binding was made. The crystal structure of OAT in complex with 5FMO (PDB: 2OAT) provides a good starting point for predicting the orientation of L-Orn into the active site of the wild-type and the R180T mutant forms of OAT. As discussed earlier [6], the 5FMO molecule differs from the Orn-PLP external aldimine only for a carbon–carbon double bond in place of the Schiff base, and a carbonyl group instead of a methylene group in the δ-position. Thus, we used the 5FMO-PLP external aldimine in a template docking approach to model the position of PLP-L-Orn in both wild-type and R180T variant. After docking, no major changes regarding the orientation of the pyridine ring and of the active site residues, as well as no stereochemical violations, were observed comparing L-Orn in wild-type and R180T OAT (Fig. 4A,B respectively). Due to the replacement of Arg180 by Thr, however, the important anchoring ion-pair interaction between the α-carboxylate of L-Orn and Arg180 is lost in the mutant enzyme, which accounts for the different predicted ΔG binding in the two dockings (L-Orn-wild-type-OAT ~ -14.27 Kcal·mol^{-1}; L-Orn-R180T-OAT ~ -12.62 Kcal·mol^{-1}). The latter interaction has been suggested to prevent the α-transamination of the substrate [6]. Therefore, Arg180 acts as a pivotal anchoring point for ensuring that only the δ-amino group can approach the Schiff base. In the absence of the positive charge of Arg180, another active site residue, namely Arg413, could bind and stabilize the negative charge of the α-carboxylate of L-Orn. Interestingly, previous studies on other aminotransferases demonstrated that homologous residues to Arg413 in OAT form an ion pair with the α-carboxylate group of the substrates [14]. Therefore, prompted by such considerations, we considered also an alternative binding mode of L-Orn to the R180T OAT mutant, i.e. an external aldimine involving the binding of PLP through the α-amino, in place of the δ-amino group (Fig. 4C). To this end, and according to the 'Glu235 switch' hypothesis of aminotransferases binding [15,16], the side chain of Glu235 was modified in order to make Arg413 available for interaction with the α-carboxylate of L-Orn. This switch can be accomplished by small rotations around the χ1 and χ2 angles of Glu235 in such a way that the carboxylate group comes into contact with bulk solvent while at the same time being hydrogen-bonded to Thr407 [6], without altering any backbone position. As shown in Fig. 4C, after molecular docking followed by energy minimization, the very reasonable geometry of binding and predicted stability (ΔGbinding ~ -11.04 Kcal·mol^{-1}) obtained for this second binding mode of L-Orn fully supports the hypothesis that, in the presence of the R180T mutation, L-Orn can be accommodated into two alternative orientations, in which both the α- and δ-amino groups compete for covalent PLP binding and formation of the external aldimine. The docking results predict that the reaction of the R180T variant with L-Orn will generate, in addition to P5C, the product of the α-transamination, i.e. P2C (Scheme 2).

NMR analysis of the L-Orn transamination

To validate the predictions of molecular docking, since the typical activity measurements do not distinguish the 2- or 5-amino group of L-Orn used in the transamination reaction catalysed by OAT, NMR

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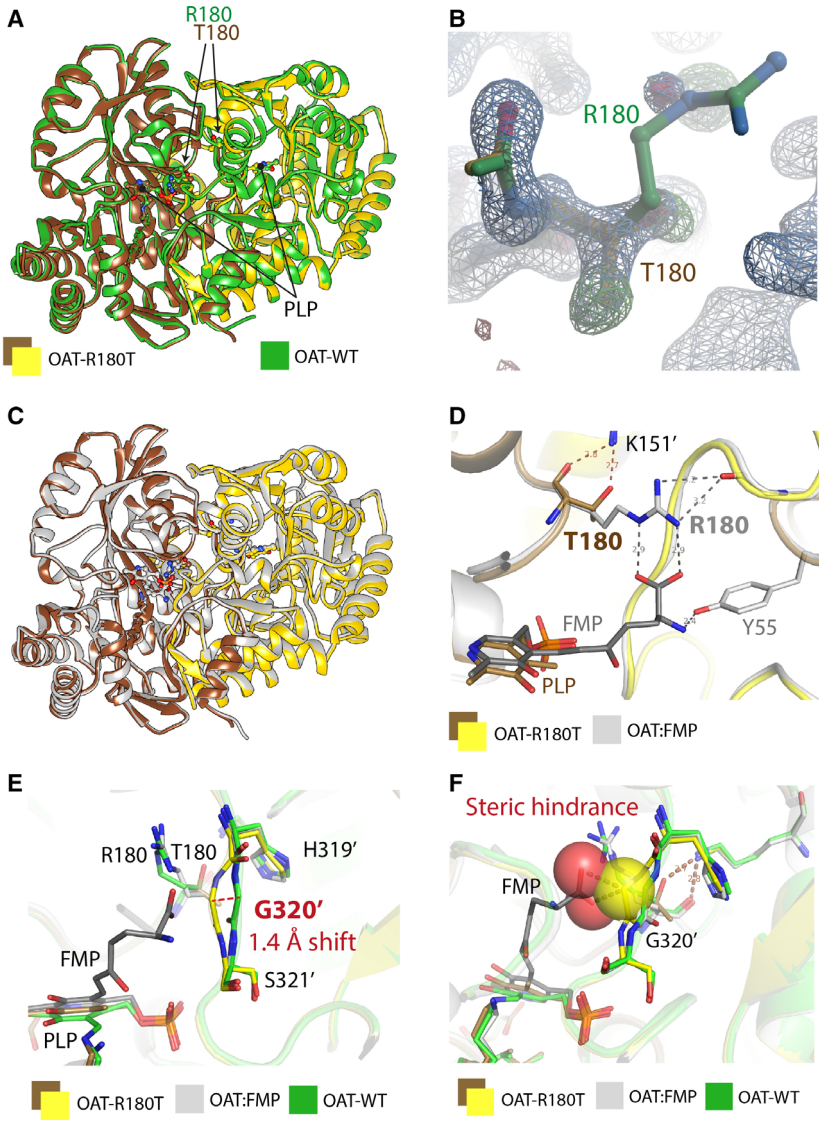


Fig. 3. Structure of the R180T mutant. (A) Structural superposition of the OAT-R180T mutant (the two chains of the dimer are shown in brown and yellow) and OAT wild-type (in green; PDB: 1OAT; [4]). The two structures superpose with a r.m.s.d. of 0.25 Å. The position of the mutated side chains and of the two bound coenzymes (PLP) is also indicated. (B) First electron density maps ($2F_o - F_c$ blue; $F_o - F_c$ green; contour level is 1.5 and 3.5 sigma respectively) obtained by molecular replacement using OAT-wt structure as search model (PDB: 1OAT). The electron density for the mutated residue (R180T) is very clear. (C) Structural superposition of the OAT-R180T mutant (in brown and yellow) and OAT in complex with the analogue of the external aldimine; 2-ammonio-7-(2-methyl-3-oxido-5-((phosphonoxy)-methyl)-4-pyridoxyl-5-oxo-6-heptenoate (FMP; in white; PDB: 2OAT; [5]). (D) Interactions of FMP (light-grey) with different residues of the active site. Dashed lines indicate the key interaction of R180, T180 and of the FMP inhibitor. (E) Superposition of the mutant (in brown and yellow), wild-type (in green; PDB: 1OAT; [4]) and FMP bound (in white; PDB: 2OAT; [5]) structures showing the different conformation of the main chain in the region of the partner subunit ^{319}HGS^{321}. The Cα of Gly320′ are shifted of 1.4 Å. (F) The clash between the shifted Gly320′ of OAT-R180T and the carboxyl group of FMP (PDB: 2OAT [5]) is highlighted using the Van der Waals radii of the atoms. It is likely that this shift would also partially hamper the correct positioning of the external aldimine, due to a steric hindrance with the carboxyl of ornithine.

spectroscopy was used to track the OAT enzymatic activity and to identify the products formed in the transamination reaction. The conversion of L-Orn into GSA and the spontaneous intramolecular cyclization to form P5C, was monitored recording various real-time mono-dimensional ^{1}H-NMR spectra after the addition of purified wild-type OAT. As shown in Fig. 5, the intensity of the substrate proton signals of L-Orn (peaks A, B, C, D) and of α-KG (peaks E, F) decreased during the course of the reaction. In parallel,

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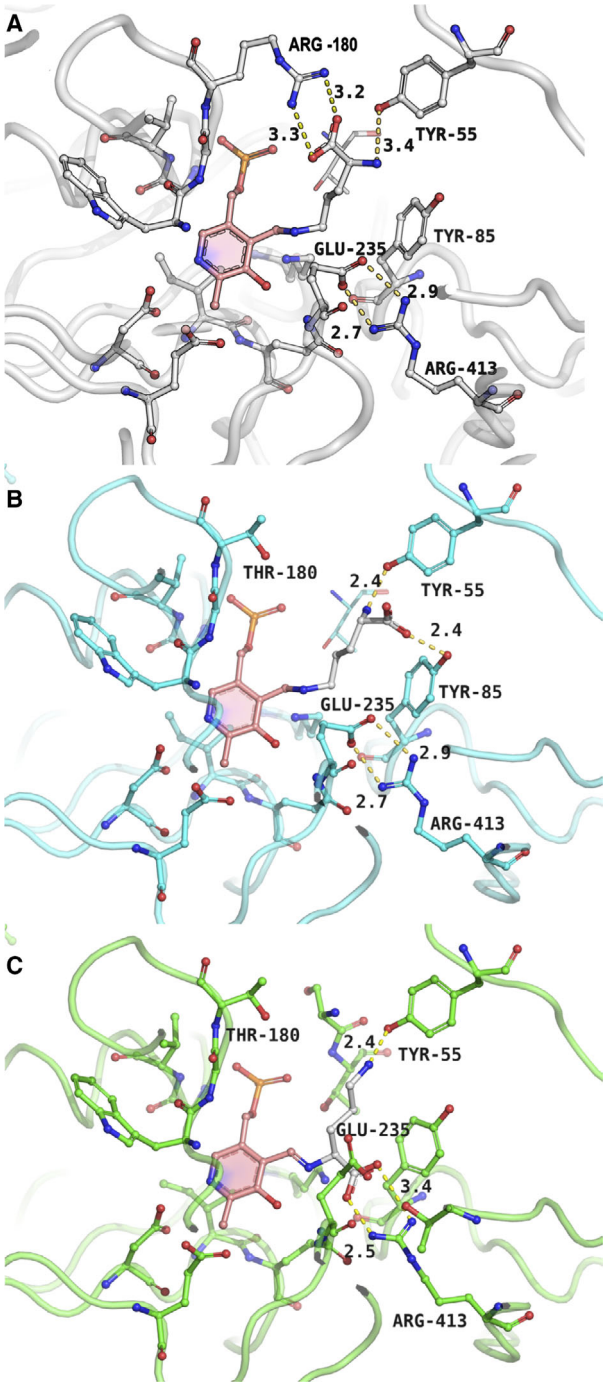
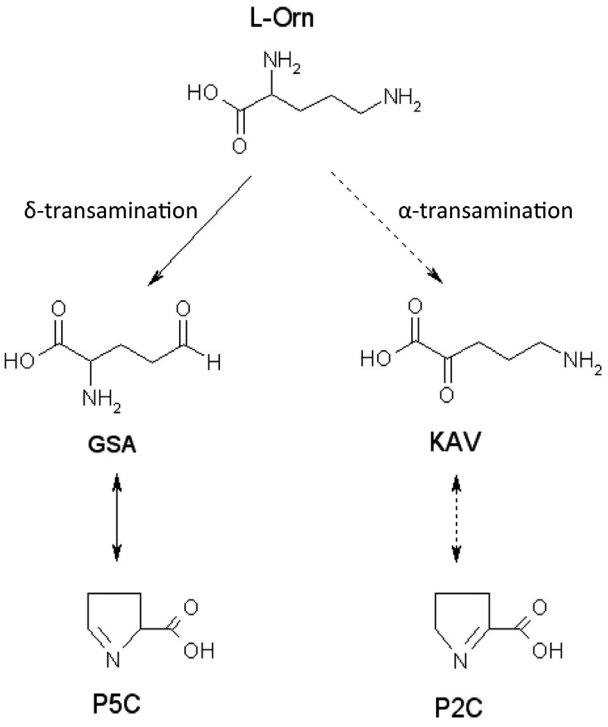


Fig. 4. Modelling of the position of PLP-L-Orn into the active site of wild-type and R180T mutant of OAT. The predicted orientation of L-Orn when bound to PLP (pink sticks) as external aldimine with its δ-amino group into the active site of wild-type OAT (A) or R180T variant (B), and with α-amino group into the R180T variant active site (C). Dashed lines labelled with distances (Å) indicate key interactions for the stabilization of L-Orn (represented as white sticks).



Scheme 2. Reaction products from δ- or α-transamination of L-Orn. GSA, Glutamic-γ-semialdehyde; KAV, α-ketoaminovalerate; P2C, Δ^1-pyrroline-2-carboxylate; P5C, Δ^1-pyrroline-5-carboxylate.

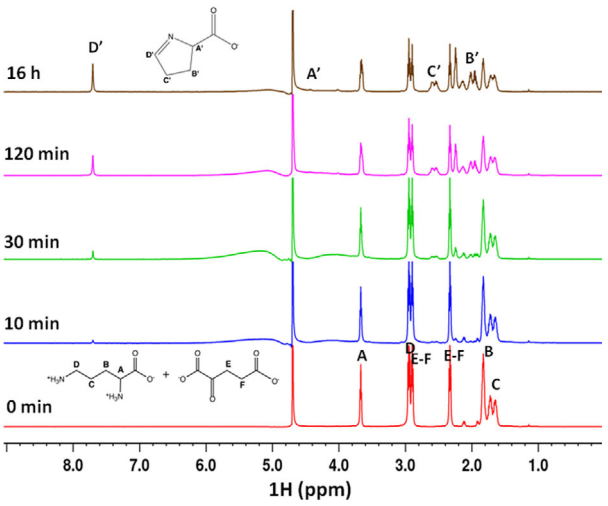


Fig. 5. NMR analysis of L-Orn transamination catalysed by OAT wild-type. ^{1}H-NMR spectra of wild-type OAT recorded as a function of time. The resonances of substrates (t = 0 min) and product (t = 16 h) are labelled on the structures.

the appearance of time-increasing signals was observed from the earliest time point up to ~ 16 h. These changes indicated that the substrates were gradually consumed and the formation of P5C increased during

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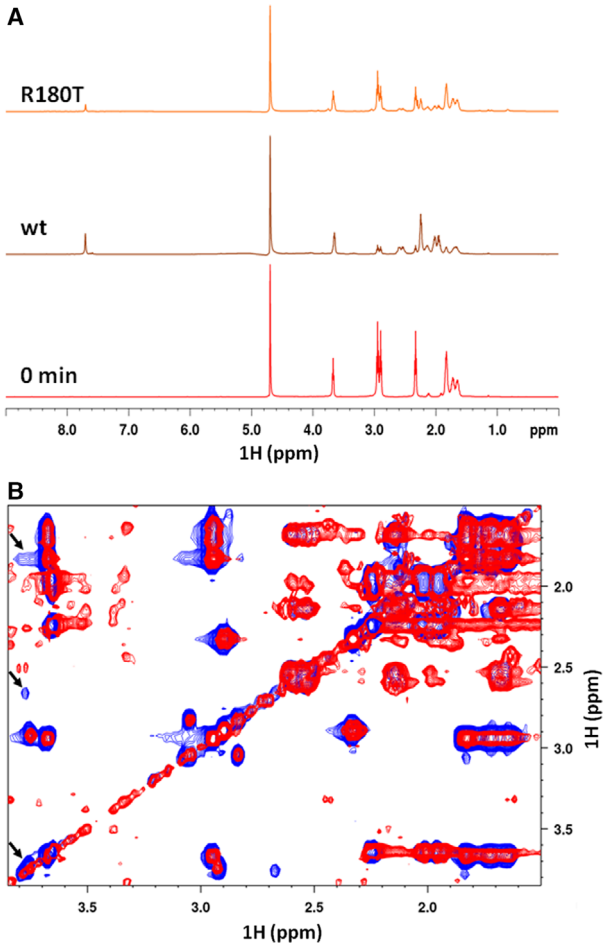


Fig. 6. NMR analysis of the reaction products of wild-type and R180T OAT variant. (A) Comparison of ^{1}H NMR spectra of the reaction mixture before (red) and after 16 h from the addition of wild-type (brown) or R180T (orange) OAT. (B) Overlay of a spectral region from 2D ^{1}H–^{1}H TOCSY recorded on the reaction mixture upon addition of wild-type (red) or R180T mutant (blue). Arrows indicate the cross-peaks appeared in presence of the mutant and assigned to P2C.

the course of the reaction (labelled A′–D′). The resonances assigned to the P5C were confirmed by the analysis of two-dimensional total correlated spectroscopy (TOCSY) spectrum. The ^{1}H-NMR experiments, acquired on the reaction mixture containing the purified R180T mutant, revealed the formation of the P5C during the course of the reaction and therefore the δ-transaminase activity of the mutant. On the other hand, the comparison of spectra collected on the wild-type and mutant at the end of the time course (Fig. 6A) showed a lower signal intensity of resonances of P5C after the incubation with the R180T variant, confirming that the mutation on this residue causes a loss of activity. However, a detail inspection of the superimposition of 2D ^{1}H–^{1}H TOCSY spectra, in which the ^{1}H-NMR overlap signals were well resolved, showed additional cross-peaks for the R180T mutant (Fig. 6B). These peaks, belonging to a spin system constituted by three proton signals at 3.77, 2.66 and 1.85 p.p.m., suggest the formation of an additional product that can be assigned to P2C. The chemical shift of the putative P2C are indeed comparable to those of a previous NMR spectrum of P2C obtained by direct chemical oxidation of L-proline [17]. These results, in line with the predictions obtained by molecular docking studies, provide evidence that, unlike for the wild-type, the reaction of the R180T mutant with L-Orn generates not only P5C (even if in an amount lower than that of wild-type) but also P2C, the product of the α-transamination. We thus provide evidence that, in addition to the so-called 'gateway system' (Arg413 and Glu325 in human OAT) [18], Arg180 contributes to the specialization of OAT for transamination of the δ-amino group.

Conclusions

To explore the molecular basis of the pathogenesis of the R180T variant associated with GA, the structural and functional features of the mutant were defined and compared with those of the wild-type. We applied an integrated strategy, which uses biochemical, computational, X-ray and NMR studies. Arg180, an interfacial dimeric residue located at the active site of OAT, interacts with the α-carboxylate of 5FMO, a substrate analogue [6]. The first point deriving from this investigation is that the R180T mutation causes an active site perturbation suggested by spectroscopic measurements and confirmed by the solved crystal structure of the mutant. As a consequence, we noticed that the R180T mutant exhibits an increase of the K_m value for L-Orn, of the coenzyme binding affinity as well as of the thermostability. The second finding is that the R180T mutant in the purified recombinant form displays a remarkable drop of the k_{cat} value for L-Orn, in agreement with what previously observed at cellular level [10]. As predicted by molecular docking studies and validated by measurements of the enzymatic activity of wild-type and mutant by NMR spectroscopy, we found that the replacement of Arg180 by Thr allows not only the δ-transamination, even if with a reduced catalytic activity with respect to the wild-type, but also α-transamination of L-Orn. This indicates that L-Orn is not able to acquire a correct positioning at the active site of the R180T mutant. It can be suggested that the loss of the interaction between the α-carboxylate of L-Orn and Arg180 and/or the altered state of the

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bound coenzyme in the mutant might be the reason(s) for the competition between the external aldimine leading to α-transamination and the one leading to δ-transamination. It cannot be also ruled out that just because of the above-mentioned molecular defects, the latter external aldimine is not properly oriented, i.e. in a conformation not completely productive for δ-transamination. Overall, these considerations provide a plausible interpretation of the molecular reasons underlying the loss of the catalytic efficiency of the R180T mutant.

Altogether, these data allow us to (a) provide evidence that Arg 180 optimizes the δ-transamination, thus preventing its α-transamination and (b) acquire a complete knowledge of the enzymatic phenotype associated with the pathogenic R180T mutation. It is of interest to analyse if the catalytic defect of R180T variant could have possible relapse(s) on L-Orn metabolism. The P2C generated by this pathogenic variant should not be relevant considering its modest production. Instead, the marked decrease of the P5C production deserves a note. It is known that the metabolic role of OAT is not only related to L-Orn degradation but also to the synthesis of proline. The latter process depends on the formation of P5C, which is reduced by P5C reductase. Recent metabolomic studies clearly indicated that proline is a preferred nutrient source for cultured retinal pigment epithelial cells [19]. Thus, it is possible to speculate that the pathogenic effect of the R180T mutation on the δ-transamination, leading to a reduced ability to catabolize L-Orn, would negatively affect proline synthesis. Another point to consider are the possible implications for the pathogenic R180T mutation in terms of therapy strategy. Pyridoxine (vitamin B6) administration is one of the most widespread pharmacological therapies for inherited diseases involving deficit of PLP-dependent enzymes. Pyridoxine is known to be converted to PLP inside the cell, and the effect of PLP is not limited to mutations that affect the apo → holo conversion, but also to those that cause folding defects. In the latter case, PLP exerts a chaperone role [20]. Although a subset of patients with GA showed to be pyridoxine responsive [12,21–23], the molecular bases underlying responsiveness was clarified for only the V332M variant. In fact, it was demonstrated that this variant, prone to release the coenzyme during catalysis, converts into the apoform which undergoes unfolding and aggregation under physiological conditions [24]. In contrast, as for the wild-type OAT, PLP remains bound to the R180T variant during catalysis. This suggests the B6-non-responsiveness of the R180T variant, in complete agreement with previous data [12].

Materials and methods

Materials

Pyridoxal 5′-phosphate, L-Orn, α-KG, 2-aminobenzaldehyde, DMSO, IPTG, phenylhydrazine hydrochloride, hydroxylamine and PMSF were purchased from Sigma (Sigma-Aldrich, St. Louis, MO, USA). All other chemicals were of the highest purity available.'

Site-directed mutagenesis

Site-directed mutagenesis was performed by means of the QuikChange II site-directed mutagenesis kit (Agilent Technologies, Santa Clara, California) using pOAT [4] as templates. The pathogenic mutation R180T was introduced by using the primer 5′-GGGAACTTCTGGGGTACGACG TTGTCTGCTATC-3′ and its complement (the mutated codon is underlined). Successful mutagenesis was verified by DNA sequencing analysis of the entire open reading frame.

Expression and purification of R180T mutant

The conditions used for expression and purification of the R180T mutant were as described for the wild-type enzyme. The protein concentration was determined by absorbance spectroscopy using a previously estimated extinction molar coefficient of 124 767 $M^{-1}\cdot cm^{-1}$ [4]. The PLP content of the mutant was determined by releasing the coenzyme in 0.1 N NaOH and using an ε_M of 6600 $M^{-1}\cdot cm^{-1}$ at 388 nm [25].

Enzyme activity assays by absorbance and NMR spectroscopy

The OAT enzymatic activity was determined by a spectrophotometric assay measuring the dihydroquinazolium derivative of P5C after incubation with 2-aminobenzaldehyde, as already described [4]. Steady-state kinetic parameters were determined as previously reported [4]. This assay was used during the purification procedure and to measure the steady-state kinetic parameters. The OAT enzymatic activity was also measured by NMR spectroscopy, which allows to monitor continuously the course of the reaction and to identify the products of the transaminase reaction. The procedure is the following: L-Orn (2 mM) and α-KG (2 mM) in 50 mM potassium phosphate buffer, pH 8 were added to a standard 5 mm NMR tube and analysed before the addition of the enzymes (representing $t = 0$ min). The enzymatic reactions were initiated upon addition of 50 nM of purified wild-type or 10 μM of R180T mutant to the NMR tube. After rapid mixing, the reaction mixtures were analysed by 1D and 2D NMR spectroscopy at different times. NMR measurements were carried out using 550 μL of solution with 10% D_2O as a locking substance. All NMR experiments were collected at 25 °C with a 600 MHz Bruker Avance III spectrometer (Karlsruhe,

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Germany) quipped with a TCI cryoprobe. The 1D ^{1}H-NMR spectra were recorded using the Bruker-standard 'zgesgp' sequence with 16 scans and a spectral width of 12019 Hz. Spectra were referenced to the suppressed water signal (4.70 p.p.m.). The 2D ^{1}H–^{1}H TOCSY spectra were performed with a 60-ms mixing time, using Bruker's 'mlevesgpph' pulse sequence. 2D spectra were acquired with four total transients, a time domain of 2k points (F2) and 512 experiments (F1) and a spectral window of 10200 Hz in both dimensions. All spectra were manually phased and baseline corrected using TOPSPIN 3.2 (Bruker). ^{1}H-NMR signals of L-Orn and α-KG were assigned based on comparisons with the chemical shifts of standard metabolites in the Biological Magnetic Resonance Data Bank (www.bmrb.wisc.edu).

Spectroscopic measurements

Absorption measurements were performed by a Jasco V-550 spectrophotometer (Jasco Europe Srl, Cremella (LC), Italy) at a protein concentration of 6 μM. CD measurements were made by a Jasco J-710 spectropolarimeter at a protein concentration of 6 μM. For near-UV and visible wavelengths, the protein concentration varied from 5 to 10 μM in a cuvette with a path length of 1 cm. Routinely, three spectra were recorded at a scan speed of 50 nm·min^{-1} with a bandwidth of 2 nm and averaged automatically. For far-UV measurements, the protein concentration was 0.1 mg·mL^{-1} with a path length of 0.1 cm. Fluorescence spectra were taken with a FP750 Jasco spectrofluorometer using 5 nm excitation and emission bandwidths at a protein concentration of 20 μM.

Thermal stability studies

The thermostability studies on the purified R180T variant were performed by CD-monitored thermal unfolding at 222 nm. CD signal was registered at 6 μM protein concentration in 50 mM Hepes plus 150 mM NaCl, pH 8, with temperature increasing from 25 to 90 °C at a heating rate of 1.5 °C·min^{-1}. The PLP or PMP dissociation from PLP- or PMP-bound enzyme, respectively, during thermal unfolding was also monitored by measuring the CD signal at 420 and 333 nm, respectively, under the same experimental conditions reported above. Calculation of the melting temperatures (T_m) was carried out by fitting the CD signal to a two state unfolding model using the ORIGIN PRO 7.0 software (OriginLab Corporation, Northampton, MA).

Crystallography and X-ray structure determination

Single crystals (70 × 70 × 200 μm) of the OAT-R180T mutant were obtained by the vapour diffusion, sitting drop, method, mixing 400 nL of a 115 μM protein solution (150 mM NaCl; 50 mM Hepes pH 8.0) with an equal volume of precipitant solution (4.0 M ammonium acetate; 0.1 M Bis-Tris propane pH 7.0). Crystals were cryoprotected by rapid soaking in mother liquor containing 20% glycerol, before flash freezing in liquid nitrogen. Diffraction data were collected on the ELETTRA-XRD1 beamline at a fixed wavelength (0.999 Å). Data processing was performed with XDS [26] and Aimless [27,28]. Initial phases were calculated with MOLREP [29] using the structure of the wild-type enzyme as template (PDB: 1OAT) [5]. Cycles of model building and refinement with COOT [30] and REFMAC5 [31] were performed. The crystals contained three molecules in the asymmetric unit with a solvent content of 44% (Matthews coef.: 2.2). Three PLP molecules were modelled in the electron density, in all chains the Schiff base linkage with Lys292 appeared to be photo reduced as previously observed for other PLP-dependent enzymes [32], and was modelled accordingly. Data collection and refinement statistics are summarized in Table 3. Coordinates and structure factors have been deposited in the protein data bank with accession code: 6HX7. Structural images were produced with the UCSF Chimera package developed by the Resource for Biocomputing, Visualization, and Informatics at the University of California,

Table 3. Data collection and refinement statistics. Values in parentheses refer to the highest resolution shell.

OAT-R180T	
Data collection	
Beamline	Elettra XRD1
Space group	$P3_11\ 2$
Cell dimension; a, *b*, *c* (Å); α, β, γ (°)	193.3, 193.3, 57.2; 90, 90, 120
Resolution range (Å)	48.3–1.8 (1.83–1.80)
Half-set correlation CC(½) (%)[a]	99.7 (94.0)
$<I/\sigma I>$	7.6 (2.4)
Completeness (%)	98.7 (96.2)
No. of reflection (total)	888 909 (26 568)
Multiplicity	8.0 (5.0)
Wilson *B*-factor (Å^2)	11.1
Refinement	
Resolution range (Å)	45.0–1.8
No. of unique reflections	105 147
R_{work}/R_{free} (%)	22.1/22.8
F_o, F_c correlation	0.96
Average *B*-factor all atoms (Å^2; no. of atoms)	16.0 (10 446)
Bonds RMSD	
Length (Å)	0.021
Angle (°)	1.473
Ramachandran (no. of residues, %)	
Favoured	1133 (96.0)
Allowed	46 (3.9)
Outliers	1 (0.1)

[a] Percentage of correlation between intensities from random half-data sets.

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San Francisco (supported by NIGMS P41-GM103311; http://www.cgl.ucsf.edu/chimera) [33].

Molecular docking

The Dundee PRODRG2 Server [34] was used to build the energy minimized three-dimensional structures of the Nα-PLP-L-Orn and Nδ-PLP-L-Orn external aldimines complexes, which were then docked into the active site of the R180T structure of OAT, using the template-based molecular docking approach of MOLEGRO VIRTUAL DOCKER (MVD) software (CLCbio®, Aarhus, Denmark). Flexible torsions of the external aldimines were automatically detected by MVD, and manually checked for consistency. A search space of 15 Å radius, centred on the active site cavity, was used for docking. PLP complexed with the highly specific inhibitor 5FMO (PDB: 2OAT) in its internal aldimine form was taken as pharmacophoric group for template-based docking [35]. In the latter, if an atom of the ligand matches a group definition, it is rewarded by using a weighted score that depends on its distance to the group centres. The grid-based MolDock score with a grid resolution of 0.30 Å was used as scoring function, and MolDock SE was used as docking algorithm [36]. For each ligand, 10 runs were defined. Similar poses (RMSD $\leq$ 1.0 Å) were clustered, and the best scoring one was taken as representative. Other docking parameters were fixed at their default values. After docking, energy optimization of hydrogen bonds was performed.

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Conflicts of interest

The authors declare no conflict of interest.

Author contributions

RM, AP, BC and CBV designed research; RM produced the enzymes and carried out its biochemical characterization; AP performed molecular docking investigation; GG and FC crystallized and solved the R180T mutant; SZ performed NMR experiments; CBV wrote the paper with contribution and approvals of all authors.

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